Path2Prevention

A Lifestyle Recommendation System for Diabetes Prevention

(Comprehensive Presentation & Technical Report)

Slide 1: Project Introduction - Detailed Problem Statement

The Global Epidemic of Type 2 Diabetes

Type 2 Diabetes (T2D) is currently one of the fastest-growing global health emergencies. According to the World Health Organization (WHO), hundreds of millions of people currently live with the condition, and millions die prematurely every year due to its complications (like cardiovascular disease, blindness, and kidney failure).

The "Invisible" Pre-Diabetes Phase

Before developing full T2D, individuals enter a stage called "pre-diabetes" where blood sugar levels are higher than normal but not high enough for a clinical diagnosis. This phase is almost entirely asymptomatic. Because there are no visible symptoms, millions of people live with high-risk lifestyle habits completely unaware of the damage occurring to their metabolic systems.

The Demographic Vulnerability (South Asian Populations)

Slide 2: Project Introduction - Deep Need of the Project

The Failure of Generic Advice

When individuals visit a doctor, they are often told universally generic advice: "Eat less, move more." This advice is notoriously ineffective because it lacks personalization. It does not account for an individual's specific daily habits, cultural dietary patterns, sleep quality, or stress levels.

The Healthcare Burden

Our modern healthcare system is inherently "reactive." Doctors treat diseases after symptoms appear. By the time a clinical diagnosis of Type 2 Diabetes is made, up to 50% of the insulin-producing beta cells in the pancreas may already be destroyed. There is a desperate need for a "proactive" system that prevents the disease before it happens.

Our Innovative Solution

Slide 3: Project Introduction - Core Objectives

Primary Objectives:

1. **Develop an Intelligent Rule Engine:** Build a backend computation system (`pipeline.js`) that translates complex medical guidelines from the World Health Organization (WHO), International Diabetes Federation (IDF), and American Diabetes Association (ADA) into logical software rules.
2. **Dynamic Risk Scoring:** Provide users with a highly accurate, daily Risk Score (scaled from 0 to 100) that mathematically adjusts based on family history, localized BMI thresholds, waist circumference, and daily behavioral deficits.
3. **Targeted Interventions:** Replace overwhelming medical advice with exactly 3 to 5 highly-prioritized, actionable daily recommendations tailored specifically to the user's weakest areas derived from their logs.

Secondary Objectives:

Slide 4: Planning of Project Work - Project Timeline (Gantt Chart)

(Represent this strictly as a visual Gantt chart in the Presentation)

- **Phase 1: Research & Blueprinting (Week 1)**
  - Requirement analysis and demographic medical research (IDF guidelines).
  - System Architecture mapping and MongoDB database schema design.
- **Phase 2: Core Backend Engine Development (Week 2)**
  - Constructing the Express.js REST API.
  - Developing the `pipeline.js` algorithm (calculating sub-scores for activity, sleep, diet, and physical measurements).
- **Phase 3: Frontend Interface & UI Generation (Week 3)**
  - Integrating Stitch-inspired Material Design UI.
  - Connecting HTML5/Vanilla JS frontend with backend JWT-secured APIs (Login/Register, Daily Logs, Profile).

Slide 5: Team Structure & Roles

(Customize individual names formatting as needed)

- **Project Manager / Team Leader:**
    - **Responsibilities:** Oversees the overall system architecture, product vision, and ensures that the software algorithms perfectly align with clinical diagnostic guidelines (WHO/IDF). Manages the timeline and agile sprint delivery.
- **Full-Stack Developer:**
    - **Responsibilities:** Writes all backend logic including Mongoose data schemas, Express.js routing, and the heavy-lifting custom Rule-Base Engine. Also responsible for writing Vanilla JS asynchronous frontend network requests (`api.js`) and state management.
- **UI/UX Designer:**
    - **Responsibilities:** Crafts the "Clinical Serenity" visual aesthetic. Selects custom

Slide 6: Design Methodology - Development Model

Model Used: Agile Methodology

Why we selected Agile over Waterfall:

1. **High Iteration Speed:** Building a complex mathematical health rule-engine requires constant mathematical tweaks. Waterfall requires perfect upfront planning, but Agile allowed the team to build feature functionality in modular, 1-week sprints (e.g., building the Log Form, then the Dashboard, then the Insights Analytics).
2. **Continuous Feedback Loop:** During development, we realized generic Western BMI thresholds were inaccurate for our target demographics. Agile's flexibility allowed us to dynamically pivot our entire database pipeline to accommodate localized South Asian IDF thresholds dynamically without delaying the project.
3. **Modular Component Architecture:** Agile allowed us to develop the frontend features (Dashboard, Insights, Settings) and test them entirely independently as

Slide 7: Design Methodology - Tools Used

- **Source Code & Version Control:** Git, GitHub (for branch handling and commit histories).
- **API Testing & Debugging:** Postman (for testing raw JSON payload responses) and Browser Chrome Developer Tools (Network/Console monitoring).
- **UI/UX Prototyping & AI Assistance:** Figma (for early UI Mockups), and Stitch MCP (AI Interface Assistance mapping advanced styling).
- **Integrated Development Environment (IDE):** Visual Studio Code (VS Code) utilizing Node extensions and Prettier formatting.
- **Documentation:** Markdown (MD) for dynamic documentation and Marp CLI for programmatic presentation generation.

Slide 8: Design Methodology - Technologies Used

1. **Frontend (Client-Side Architecture):**

- **HTML5 / CSS3:** Utilizes extensive custom CSS variables (Root tokens), Flexbox/Grid for layout, and advanced Glassmorphism design aesthetics (blurred backdrops and gentle shadows).
- **Vanilla JavaScript (ES6+):** Pure DOM manipulation and asynchronous network fetching (`async/await`) completely independent of heavy frameworks like React or Angular to ensure maximum performance and lightweight serving.
- **Chart.js / HTML Canvas:** For rendering highly-responsive, multi-dataset chronological line charts and visual analytic patterns.

2. **Backend (Server-Side Architecture):**

- **Node.js & Express.js:** Fast, non-blocking, event-driven HTTP framework acting as the core traffic router for RESTful endpoints.

Slide 9: System Architecture - Block Diagram

(Visual Representation of the Data Flow between Components)

1. User Interaction (PWA Browser/Mobile):
The user accesses the application and inputs daily habits via secure HTTPS.
↓

2. Frontend Interface (HTML, CSS, JS, Chart.js):
The UI layer captures the data and routes it securely to the system logic utilizing JSON Web Tokens (JWT).
↓

3. Backend Processing (Node.js + Express.js API):
The core traffic router authenticates the request, processes the payload, and delegates it to the algorithmic `pipeline.js` component.
↓ (Read/Write)

4. Database (MongoDB Collections):

Slide 10: Conceptual Demonstration - How the System Actually Works

Step 1: Onboarding Baseline

A user creates a highly secure account using JWT authentication. They complete a 4-step onboarding assessment gathering baseline clinical markers: height, weight, biological sex, and crucial family history of diabetes—which establishes their baseline genetic multiplier.

Step 2: Micro-Habit Logging

Instead of counting every single calorie (which causes user burnout), the system relies purely on "Micro-Habits". The user utilizes minimalist sliders and toggles to submit a daily log covering steps, sleep duration, sugary drink intake, fast food consumption, and subjective stress levels.

Step 3: Algorithmic Pipeline Calculation

Slide 11: Conceptual Demonstration - Flowchart of the Process

Step 1. START: User Registration / Login

The user securely enters the application.

Step 2. CONDITION: Is Onboarding Complete?

- *If No:* Route to Onboarding Diagnostic Survey to save the initial Profile.
- *If Yes:* Route directly to the main Dashboard Landing UI.

Step 3. ACTION: Daily Submission

User submits their Daily Habit Log Form tracking their micro-habits.

Step 4. PROCESS: Backend Validates Data

The server sanitizes the input and enforces a log lock to ensure data immutability.

Step 5. TRIGGER: Execution of Rule Engine

The system triggers the `pipeline.js` algorithm to calculate global metabolic deficits.